LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 12 Test (Set B): Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

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SECTION A

Q1. Outside a bar magnet, magnetic field lines go: **[1]**

Answer: (a) from the north pole to the south pole

Q2. A current-carrying conductor placed parallel to a magnetic field experiences: **[1]**

Answer: (d) no force

Q3. Which instrument detects the presence of a current in a circuit? **[1]**

Answer: (b) Galvanometer

Q4. Which of the following increases the strength of an electromagnet? **[1]**

Answer: (b) Using a soft iron core

Q5. Which safety device works on the heating effect of current? **[1]**

Answer: (d) Fuse

Q6. **Assertion (A):** Appliances with metal bodies are connected to the earth wire. **Reason (R):** The earth wire carries the main current to the appliance. **[1]**

Answer: (c). The earth wire is a safety wire. It carries leaked current to the earth, not the main current.

SECTION B

Q7. Describe the magnetic field lines of a current-carrying circular loop. Why is the field strong at the centre? **[2]**

Answer: Near the wire, the field lines are concentric circles, which grow larger farther from the wire. At the centre of the loop they appear as nearly straight lines. Every part of the loop produces a field in the same direction at the centre, so the fields add up and make it strong there.

- Pattern: 1
- Fields add at centre: 1

OR, answer: A compass needle is itself a small magnet. Its poles are attracted or repelled by the poles of the bar magnet, which exerts a force through its magnetic field, so the needle turns.

- Needle is a small magnet: 1
- Force of the magnet's field turns it: 1

Q8. Give two precautions that help avoid overloading of domestic circuits. **[2]**

Answer: (Any two) Do not connect too many appliances to a single socket. Use high-power appliances only on the 15 A circuit. Use a fuse or MCB of the correct rating. Replace damaged wires and plugs.

- Any two: 1 + 1

SECTION C

Q9. A circular loop of wire lies flat on a table, and current flows through it clockwise (as seen from above). **[3]**

(i) If you hold a current-carrying conductor in your right hand with the thumb along the current, your curled fingers show the direction of the magnetic field lines. [1]
(ii) Perpendicular to the table, pointing into it (downwards). [1]
(iii) Perpendicular to the table, pointing out of it (upwards). [1]

Q10. Give three ways to increase the force on a current-carrying conductor placed in a magnetic field. **[3]**

Answer: (1) Increase the current through the conductor. (2) Use a stronger magnet (stronger magnetic field). (3) Increase the length of the conductor that lies in the field, keeping it perpendicular to the field.

- Each way: 1

OR, answer: (i) Bulb: I = P/V = 100/220 ≈ 0.45 A, which is less than 5 A, so it is safe. (ii) Heater: I = 2000/220 ≈ 9.1 A, which is more than 5 A, so it is not safe. It would overload the circuit and blow the fuse, so it should be used on the 15 A circuit.

- Bulb: 1
- Heater: 1
- Conclusion: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Hang a small aluminium rod horizontally from two connecting wires. Place a strong horseshoe magnet so that the rod lies between its poles, with the field perpendicular to the rod. Connect the rod to a battery, a key and a rheostat. When the key is closed, the rod is pushed sideways (displaced). Reversing the current pushes it the opposite way. [3]

- Set-up: 1
- Observation (rod displaced): 1
- Reversal shows direction depends on current: 1

(ii) The rod is displaced in the same direction as before. Reversing one of them reverses the force, so reversing both brings the force back to its original direction (Fleming's left-hand rule). [2]

- Same direction: 1
- Reason: 1

OR, answer: (a) Pass a thick copper wire vertically through a horizontal cardboard sheet and sprinkle iron filings on the sheet. Pass a current and tap the cardboard: the filings arrange themselves in concentric circles around the wire. A compass placed on a circle shows the direction of the field, which reverses when the current is reversed. (b) A larger current makes the field stronger. Moving farther from the wire makes the field weaker.

- (a) set-up 1, concentric circles 1, direction with compass 1
- (b) 1 + 1

SECTION E

Q12. An electric current always produces a magnetic field. Even weak ion currents travelling along the nerve cells in our body produce magnetic fields.… **[4]**

(i) Weak ion currents travelling along nerve cells. [1]

(ii) The heart and the brain. [1]

(iii) MRI (Magnetic Resonance Imaging) is a technique that uses magnetic fields to produce images of internal organs. It helps doctors find problems inside the body without surgery. [2]

- What MRI is: 1
- Usefulness: 1